

Santhosh Narayanan Baburaman

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EDUCATION

M.S. Analytics, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026

Coursework: Data Engineering, Enterprise Cloud Analytics, Predictive Analytics, Decision Analysis

B.Tech. Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

Publication: Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

EXPERIENCE

Data Analyst Intern | Philadelphia Department of Aviation Jun 2026 – Aug 2026

- Designed and owned 2 production data pipelines covering ~30% of airport operating revenue, replacing ~60 hours/month of manual review with automated, reusable workflows
- Built a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of flight-level billing data spanning 4 fiscal years; standardized airline, aircraft, and timestamp fields with fallback logic for missing or malformed source values
- Rebuilt a second pipeline on PropWorks SQL → Amazon Redshift, correcting aggregation-order and duplicate-rate defects upstream of reporting, and modeled fiscal-year time intelligence and a statistical anomaly layer in Power BI/DAX
- Integrated both pipelines to compute cross-domain passenger capture rates, and documented each workflow so the Finance team can run it independently every reporting cycle

Founder & President | ISE Analytics Student Association (IASA), USC Feb 2025 – Present

- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals

PROJECTS

Consulting Firm Data Warehouse (dbt, Snowflake, SQL, Python)

- Built a multi-layer ELT pipeline in dbt on Snowflake, transforming 17 normalized source tables into 3 subject-area datamarts for a 4-person team
- Modeled 7 conformed dimensions including an SCD Type 2 dimension preserving salary and title history, plus a generated date spine for consistent time aggregation
- Wrote 39 automated dbt tests (unique, not null, accepted values) and reconciled design docs against the production schema, documenting 13 discrepancies and unblocking 3 teammates' fact-table work within 24 hours

Automated Sales Analytics Copilot (Python, PostgreSQL, GitHub Actions, Streamlit) 🐙 GitHub

- Designed a PostgreSQL star schema with 5 KPI views tracking revenue, margin, and discount impact across a 10K-order, \$2.3M dataset
- Automated nightly inference and anomaly alerting through a GitHub Actions CI/CD pipeline, deploying a full-stack Streamlit app with Slack and email notifications

Healthcare Readmissions Pipeline (GCP Vertex AI, Kubeflow)

- Owned the data preparation stage of an 8-component Kubeflow pipeline on Vertex AI, producing versioned preprocessing artifacts consumed by training, evaluation, and endpoint-serving stages downstream

Financial Analytics & Power BI Dashboard (Python, PostgreSQL, Power BI) 🐙 GitHub

- Built an ETL pipeline extracting 10-K/10-Q data via SEC EDGAR into a PostgreSQL star schema, surfaced in a Power BI dashboard spanning 2010–2025

IPL 2026 Playoff Tracker (Python, GitHub Pages, Claude API) 🐙 GitHub

- Shipped a self-refreshing web app that pulls live match results and recomputes a 5,000-run Monte Carlo simulation on each refresh, deployed as a build-free static site

SKILLS

- **Data Engineering:** Python, SQL, dbt, ELT/ETL, Dimensional Modeling, SCD Type 2, Data Quality Testing, Parquet
- **Platforms:** Snowflake, PostgreSQL, Amazon Redshift, Alteryx, GCP (Vertex AI, Kubeflow), Azure
- **Orchestration & Ops:** GitHub Actions (CI/CD), Kubeflow Pipelines, Git
- **BI & Reporting:** Power BI (DAX, Power Query), Tableau, Streamlit, Excel